

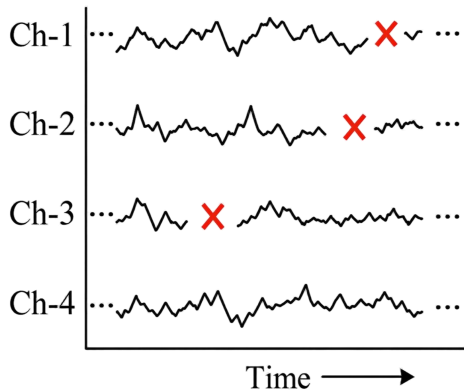


Request-Local Traffic Window (336 h×4)

Current target cell only. Traffic X. Observation mask M

No other cells or cross-request traffic cache

Structured Seasonal Evidence

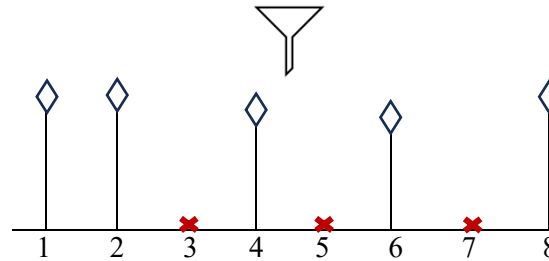


- D-1
- W-1
- W-2
- Median-7
- Median-14
- Trend
- Window
- Frozen Prior

Expert state: $\{v_j, m_j, \rho_j, \Delta_j\}$

Local context: mean, median, sd, missing rate, trend

Hard-Masked Sparse Router



$$J = \{j: m_j = 1\}$$

$$\alpha_j = \text{Entmax}_{1.5}(a_j)$$

$$j \notin J \Rightarrow \alpha_j = 0$$

Horizon and indicator conditioned query $q(h, q)$

Bounded Forecast Composition

Dynamic Baseline

$$b = \sum_j \alpha_j v_j$$

Bounded Residual

$$\delta = \delta_{\max} \tanh g_R(\cdot), \quad |\delta| \leq \delta_{\max}$$

Final Forecast

$$\hat{y} = \max \{10^{-4}, \exp(b + \delta) - 1\}$$

Training Only: fixed 15% mixed missingness augmentation + masked L1 loss



24×4 Traffic Forecast + Semantic Audit Record

$$\{v, m, \rho, \alpha, b, \delta, H, |\text{supp}(\alpha)|\}$$